

Resume

Name: Lei Ma

Position: Post-Doctoral Fellow

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PROFESSIONAL EXPERIENCE

Post-Doctoral Fellow

01/09/2024 - Present

University of Toronto, Canada

School of Civil and Mineral Engineering

Advisor: Professor Daman Panesar

Research title 1: Carbonation-activated microstructural refinement in GUL-GGBFS blended mortars: shrinkage mitigation and strength enhancement

Research title 2: Effect of GGBFS content on the durability of concrete exposed to freeze-thaw and salt scaling

Post-Graduate Intern

10/04/2023 - 29/09/2023

Holcim Innovation Centre, France

Advisor: Dr Qing Zhang

Research title: Assessment of the feasibility of applying the ring test to evaluate the cracking properties of 3D printed concrete

EDUCATION

PhD in Civil Engineering

01/09/2019 - 13/03/2024

Southeast University, China

School of Materials Science and Engineering, Civil Engineering Materials

Advisor: Professor Yamei Zhang

Research title: Study on the impact and mechanisms of early drying environments on mechanical and shrinkage properties of 3D printed concrete

MSc in Civil Engineering

01/09/2016 - 30/06/2019

Xinjiang University, China

School of Architectural Engineering, Civil Engineering

Advisor: Professor Yong Wen

Research title: Optimisation on the sulphate resistance of grouted mortar for semi-flexible pavement

BSc in Civil Engineering

01/09/2012 - 30/06/2016

Xinjiang University, China

School of Architectural Engineering, Civil Engineering

Advisor: Professor Ya'ni Qi

Research title: Exploration into improving the properties of warm mix asphalt concrete

LIST OF PUBLICATIONS

Journal Articles

1. **Ma, L.**, Panesar, DK. (2026). Carbonation-activated microstructural refinement in GUL-GGBFS blended mortars: Shrinkage mitigation and strength enhancement. *Cement and Concrete Composites*, 165, 106347, <https://doi.org/10.1016/j.cemconcomp.2025.106347>.
2. Chang, X., Wang, X., Lu, M., Wei, X., Zhao, H., **Ma, L.**, Ingham, J., Wu, Z., Noori, M. (2025). Flexural behaviour of corrugated BFRP-concrete composite structures for tunnel lining. *Structures*, 80, 109952, <https://doi.org/10.1016/j.istruc.2025.109952>.
3. **Ma, L.**, Jia, Z., Chen, Y., Jiang, Y., Huet, B., Delaplace, A., Zhang, Y., Zhang, Q. (2024). Water loss and shrinkage prediction in 3D printed concrete with varying w/c and specimen sizes. *Cement and Concrete Composites*, 149, 105523, <https://doi.org/10.1016/j.cemconcomp.2024.105523>.
4. Niu, G., Liu, C., Jia, L., **Ma, L.**, Zhang, Y., et al. (2024), Preparation and performance analysis of 3D printed lightweight EPS concrete: Insights from the excess paste theory. *Cement and Concrete Composites*, 149, 105509, <https://doi.org/10.1016/j.cemconcomp.2024.105509>.
5. Jia, Z., Zhou, M., Chen, Y., Wang, W., **Ma, L.**, Chen, Y., Liu, C., Zhang, Y. (2024). Effect of steel fiber shape and content on printability, microstructure and mechanical properties of 3D printable high strength cementitious materials. *Case Studies in Construction Materials*, 20, 03080, <https://doi.org/10.1016/j.cscm.2024.e03080>.
6. Jia, Z., Kong, L., Jia, L., **Ma, L.**, Chen, Y., Zhang, Y. (2024). Printability and mechanical properties of 3D printing ultra-high performance concrete incorporating limestone powder. *Construction and Building Materials*, 426, 136195, <https://doi.org/10.1016/j.conbuildmat.2024.136195>.
7. Shi, Y., Jia, L., Jia, Z., **Ma, L.**, Zhang, Z., Xia, K., Chen, Y., Zhang, Y. (2024), Early-age inhomogeneous deformation of 3D printed concrete: Characteristics and influences of superplasticizer and water-binder ratio. *Journal of Building Engineering*, 86, 108956, <https://doi.org/10.1016/j.jobbe.2024.108956>.
8. **Ma, L.**, Zhang, Q., Jia, Z., Liu, C., Deng, Z., Zhang, Y. (2022). Effect of drying environment on mechanical properties, internal RH and pore structure of 3D printed concrete. *Construction and Building Materials*, 315, 125731, <https://doi.org/10.1016/j.conbuildmat.2021.125731>.
9. **Ma, L.**, Zhang, Q., Lombois-Burger, H., Jia, Z., Zhang, Z., Niu, G., Zhang, Y. (2022). Pore structure, internal relative humidity, and fiber orientation of 3D printed concrete with polypropylene fiber and their relation with shrinkage. *Journal of Building Engineering*, 61, 105250, <https://doi.org/10.1016/j.jobbe.2022.105250>.
10. Zhou, W., Zhang, Y., **Ma, L.**, Li, V. (2022). Influence of printing parameters on 3D printing engineered cementitious composites (3DP-ECC). *Cement and Concrete Composites*,

130,104562, <https://doi.org/10.1016/j.cemconcomp.2022.104562>.

11. Deng, Z., Jia, Z., Zhang, C., Wang, Z., Jia, L., **Ma, L.**, Wang, X., Zhang, Y. (2022). 3D printing lightweight aggregate concrete prepared with shell-packing-aggregate method - Printability, mechanical properties and pore structure. *Journal of Building Engineering*, 62, 105404, <https://doi.org/10.1016/j.jobbe.2022.105404>.
12. Liu, C., Xiong, Y., Chen, Y., Jia, L., **Ma, L.**, Deng, Z., Wang, Z., Chen, C., Banthia, N., Zhang, Y. (2022). Effect of sulfoaluminate cement on fresh and hardened properties of 3D printing foamed concrete. *Composites Part B: Engineering*, 232, 109619, <https://doi.org/10.1016/j.compositesb.2022.109619>.
13. Liu, C., Chen, Y., Xiong, Y., Jia, L., **Ma, L.**, Wang, X., Chen, C., Banthia, N., Zhang, Y. (2022). Influence of HPMC and SF on buildability of 3D printing foam concrete: From water state and flocculation point of view. *Composites Part B: Engineering*, 242, 110075, <https://doi.org/10.1016/j.compositesb.2022.110075>.
14. Chen, Y., Jia, L., Liu, C., Zhang, Z., **Ma, L.**, Chen, C., Banthia, N., Yamei Zhang. (2022). Mechanical anisotropy evolution of 3D-printed alkali-activated materials with different GGBFS/FA combinations. *Journal of Building Engineering*, 50, 104126, <https://doi.org/10.1016/j.jobbe.2022.104126>.
15. Wang, X., Jia, L., Jia, Z., Zhang, C., Chen, Y., **Ma, L.**, Wang, Z., Deng, Z., Banthia, N., Yamei Zhang. (2022). Optimization of 3D printing concrete with coarse aggregate via proper mix design and printing process. *Journal of Building Engineering*, 56, 104745, <https://doi.org/10.1016/j.jobbe.2022.104745>.
16. Liu, C., Chen, Y., Zhang, Z., Niu, G., Xiong, Y., **Ma, L.**, Fu, Q., Chen, C., Banthia, N., Zhang, Y. (2022). Study of the influence of sand on rheological properties, bubble features and buildability of fresh foamed concrete for 3D printing. *Construction and Building Materials*, 356, 129292, <https://doi.org/10.1016/j.conbuildmat.2022.129292>.
17. Liu, C., Wang, X., Chen, Y., Zhang, C., **Ma, L.**, Deng, Z., Chen, C., Zhang, Y., Pan, J., Banthia, N. (2021). Influence of hydroxypropyl methylcellulose and silica fume on stability, rheological properties, and printability of 3D printing foam concrete. *Cement and Concrete Composites*, 122, 104158, <https://doi.org/10.1016/j.cemconcomp.2021.104158>.

Conferences Presentations

1. **Ma, L.**, Panesar, D. Is accelerated carbonation a benefit or burden to cementitious materials? A look at volume stability. *Embodied GHG Symposium*, Toronto, Canada (05/2025).
2. **Ma, L.**, Zhang, Q., Zhang, Y. The shrinkage of 3DPC, effect of surface to volume ratio, water to binder ratio, and PP fiber. *RILEM Annual Week & 1st Interdisciplinary Symposium on Smart & Sustainable Infrastructures*, Vancouver, Canada (09/2023).
3. **Ma, L.**, Zhang, Q., Lombois-Burger, H., Zhang, Y. Effect of drying environment on mechanical properties, internal RH and pore structure of 3D printed concrete. *Third International Conference on 3D Construction Printing & International Conference on 3D Printing Concrete Materials and Structures*, Shanghai, China (06/2022).

Patent

1. 3D-Printed Wall Structure for a House, 3D-Printed House, and Method for Printing. Approved by China National Intellectual Property Administration, 2023.

AWARDS

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| 1. 3D Printing Housing Hack Challenge, Holcim Innovation Centre | 01/2022 |
| 2. Design Innovation Award | 06/2021 |
| 3. “Merit Students” Award | 04/2019 |
| 4. “Outstanding Graduates” Award | 01/2019 |
| 5. Master's National Scholarship, Ministry of Education | 11/2018 |
| 6. “The Surveying Competition” prize, Xinjiang University | 04/2013 |